AI-Powered Research Gap Identifier & Trend Analyzer

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Manual research proposal evaluation is subjective and increasingly impractical as scientific literature expands. Existing AI discovery tools operate retrospectively on published corpora and cannot perform predictive, idea-level validation of unverified proposals. This paper presents a five-stage Similarity Guided Semantic Gap Detection Engine that addresses this gap. The system employs context-aware section isolation to extract gap-relevant passages (Limitations, Future Work) from retrieved literature, maps them into a 384-dimensional dense vector space via all-MiniLM-L6-v2, and constructs a blended degree-centrality graph. A few-shot prompted LLaMA 3.2 model classifies the proposal as GAP_ADDRESSING, PARTIAL_OVERLAP, or HIGH_SIMILARITY, accompanied by a grounded confidence score derived from LLM output and signal spread. Evaluated on a 12-scenario benchmark, the full system achieves 58.3% accuracy (LLaMA 3.2) and 41.7%-58.3% with a Groq cloud backend (OpenAI GPT-OSS 20B). An ablation study confirms that removing the LLM collapses accuracy to 33.3%, demonstrating the reasoning stage's criticality. The system generates grounded, explainable justifications that neither a keyword-based baseline (25%) nor a heuristic cosine-threshold approaches produce. Classification completes in under 23 seconds on consumer-grade hardware, providing a practical, human-in-the-loop decision support tool for early-stage academic research governance.

Keywords: Semantic Gap Detection, Natural Language Processing, Large Language Models, Semantic Similarity Analysis, Research Idea Validation.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
FAISS	Facebook AI Similarity Search
FR / NFR	Functional / Non-Functional Requirement
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
LLM	Large Language Model
NLP	Natural Language Processing
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
SBERT	Sentence-BERT
SDG	Sustainable Development Goal
TAF	Topic Assessment Form
TF-IDF	Term Frequency–Inverse Document Frequency

1 INTRODUCTION

1.1 Overall project context

The rapid growth of scientific publications makes it increasingly difficult for researchers, especially students and early-stage scholars, to determine whether a proposed idea addresses a genuine unresolved problem or duplicates existing work. The AI-Powered Research Gap Identifier & Trend Analyzer (Group R26-SE-023) is a collaborative research-support platform that addresses this problem through several cooperating components: a Semantic Similarity module that retrieves related literature, an Integrity & Governance module, a Recommendation & Collaboration module, a Collaborative Workspace & Progress Tracking module, and the Semantic Gap Detection Engine presented in this thesis. A web interface submits proposals to a core application server, which routes requests to these components, stores results, and presents the final report to the user (Figure 4.1).

Within this platform, the Semantic Gap Detection Engine performs the deepest analysis. It takes a user-submitted research idea, examines the limitations and future-work evidence of the most closely related papers, and produces an interpretable classification of how the idea relates to existing work. The other components are developed by other group members; this thesis covers only the design, implementation and evaluation of the engine.

1.2 Individual Component Background and Context

The growth of scientific publications has made it increasingly challenging for researchers to identify meaningful research gaps and establish the novelty of proposed research ideas. Researchers must examine existing studies, understand their contributions and limitations, and determine whether an idea addresses an unresolved problem or substantially overlaps with previous work. This process can be particularly demanding for undergraduate students and early-stage researchers who may have limited experience conducting comprehensive literature reviews.

Academic discovery platforms and literature-search systems help researchers locate related publications and explore connections between existing studies. However, finding publications that are similar to a proposed research idea does not necessarily establish whether that idea addresses an unresolved research problem. A meaningful assessment requires examining the evidence reported in relevant studies, including their limitations, methodological constraints, and recommendations for future research.

Research-gap identification therefore involves more than measuring textual similarity. A proposed idea may be highly similar to an existing paper because it investigates the same research domain while addressing a different limitation. Conversely, an idea with relatively low textual similarity may still duplicate an existing contribution at a conceptual level. An effective assessment mechanism should consequently consider

the semantic relationships between a proposed idea and evidence extracted from relevant literature, rather than relying exclusively on keyword matching or a single similarity score.

This research addresses the problem through a Similarity-Guided Semantic Analysis and Gap Detection Engine. The component forms part of a broader AI-powered research support system and is intended to assist in assessing whether a proposed research idea addresses a potential research gap, partially overlaps with previous work, or exhibits substantial similarity to existing research. Its output is intended to support academic evaluation rather than replace the judgement of researchers or supervisors.

1.3 Problem statement

Existing literature-search and research-assistance approaches can help users discover relevant publications, retrieve information, and summarize research findings. However, identifying relevant literature and validating a proposed research idea are distinct tasks. A list of related papers does not, by itself, explain whether the proposed idea addresses an unresolved limitation identified in those papers.

Manually determining this relationship requires researchers to review multiple publications, identify relevant evidence, compare research objectives and limitations, and synthesize the findings into a defensible assessment. This process can be time-consuming and may produce inconsistent judgements, particularly when the literature is extensive or when researchers are inexperienced in evaluating novelty.

The problem addressed by this research is the lack of an integrated, evidence-oriented mechanism for comparing an early-stage research idea against relevant academic literature and producing an interpretable assessment of its potential relationship to existing research. The challenge is to combine literature retrieval, targeted evidence extraction, semantic representation, similarity analysis, and language-model reasoning while limiting the influence of irrelevant context and unsupported conclusions.

Accordingly, this research investigates the development and evaluation of a similarity-guided semantic analysis engine that uses evidence from related publications to support research-gap assessment.

1.4 Research Gap

Substantial progress has been made on individual tasks relevant to literature analysis: scholarly retrieval and literature graphs [1], domain-specific and sentence-level embeddings [2], [3], topic modelling [10], [14], and automated research-gap extraction [12]. Each of these operates largely in isolation. Discovery platforms such as Semantic Scholar and Google Scholar retrieve related papers but do not assess whether a new idea addresses an unresolved limitation in them. Topic-modelling approaches identify themes and trends at corpus level. Gap-extraction systems highlight unexplored areas of a field but do not compare those areas with a specific, user-submitted idea.

Generative research assistants such as Elicit and Consensus synthesise literature to answer questions but are not designed to critique the novelty of a new hypothesis.

As a result, existing approaches are largely retrospective: they analyse published content rather than validating an early-stage idea against the closest relevant evidence. Researchers and supervisors must still read multiple papers manually, identify their limitations, and judge originality themselves. This thesis addresses that gap by developing an integrated, idea-level pipeline that combines similarity-guided retrieval, targeted extraction of limitation and future-work evidence, asymmetric semantic alignment, and LLM-based reasoning to classify whether a proposed idea addresses a gap, partially overlaps with, or largely duplicates existing research.

1.5 Aim and objectives

1.5.1 Aim

The aim of this research is to design, implement, and evaluate a Similarity-Guided Semantic Analysis and Gap Detection Engine that combines semantic analysis and Large Language Model (LLM) reasoning to assess the relationship between an early-stage research idea and relevant academic literature.

1.5.2 Objectives

The research is guided by the following objectives:

1. **Evidence extraction:** Extract limitation, future-work, methodology and discussion passages from the most relevant retrieved papers using a fallback PDF-parsing pipeline and context-aware section isolation. *Evidence: Sections 4.4.2, 5.2.1 and Table 6.x.*
2. **Semantic representation:** Represent the proposal and the extracted evidence as 384-dimensional embeddings using all-MiniLM-L6-v2. *Evidence: Sections 4.4.3, 5.2.2.*
3. **Semantic alignment:** Measure proposal-to-evidence alignment with cosine similarity and a blended degree-centrality score to expose overlap and potential gaps. *Evidence: Sections 3.3, 4.4.4.*
4. **LLM reasoning:** Use a few-shot prompted LLM to classify each proposal as GAP_ADDRESSING, PARTIAL_OVERLAP or HIGH_SIMILARITY with an evidence-grounded explanation and a blended confidence score. *Evidence: Sections 4.4.5, 5.2.3.*
5. **Evaluation:** Evaluate the component using a labelled 12-scenario benchmark (accuracy, precision, recall and F1, confusion matrix), TF-IDF and naive-RAG baselines, ablation experiments, end-to-end and stage-level latency, and expert assessment. *Evidence: Chapter 6.*

1.6 Research Questions or hypotheses

The central research question is: How can early-stage research gap detection be improved by using similarity-guided semantic analysis to determine whether a newly proposed idea addresses a genuine unresolved gap in existing literature? Which raises the following questions:

RQ1: How can similarity-guided literature retrieval and targeted evidence extraction support the identification of research gaps in early-stage research proposals?

RQ2: To what extent can semantic representations and alignment signals, combined with Large Language Model reasoning, classify the relationship between a proposed research idea and existing literature?

RQ3: How does the proposed approach perform against simpler baseline methods and alternative configurations in terms of classification accuracy and processing time?

RQ4: What limitations affect the reliability, interpretability, and practical use of automated research-gap assessments?

These questions will guide the system design, experimental evaluation, and critical discussion.

1.7 Scope limitations and assumptions (Scope of the Research)

This research focuses on the individual Semantic Analysis and Gap Detection Engine within a larger research support platform. Its central task is to analyse a proposed research idea in relation to relevant academic literature and provide an evidence-oriented assessment of possible research gaps and overlap.

The five-stage pipeline described in the research paper consists of:

1. Similarity-guided retrieval and preprocessing of relevant literature.
2. Extraction of evidence from sections associated with research contributions, limitations, and future work.
3. Generation of semantic representations using the `all-MiniLM-L6-v2` embedding model.
4. Semantic alignment and graph-based analysis of relationships between the proposal and retrieved literature.
5. Language-model reasoning to produce a structured classification and supporting explanation.

The system uses three principal classifications: `GAP_ADDRESSING`, `PARTIAL_OVERLAP`, and `HIGH_SIMILARITY`. These classifications represent decision-support outputs and should not be interpreted as definitive proof of originality or duplication.

The research paper reports an evaluation based on 12 manually curated scenarios, including comparisons with baseline approaches and an ablation without language-model reasoning. These results provide the initial evaluation baseline for this thesis. Their limitations, including the small benchmark size and the extent to which the results generalize to other research domains, will be considered in the evaluation and discussion chapters.

Experiments involving alternative embedding models, different hosted language models, provider-routing configurations, and additional implementation changes are outside the established results baseline unless they are completed and evaluated sufficiently to support reliable reporting. Such experiments may be incorporated as additional work if their configurations, test data, and outcomes can be documented and verified.

1.8 Individual Contribution and Significance of the Research

The individual component contributes a semantic assessment process to the broader research-support platform. Its technical focus is the combination of literature evidence extraction, semantic representation, alignment analysis, and LLM reasoning to support the evaluation of early-stage research ideas.

The intended contribution is to provide a more structured basis for examining a proposed idea than a list of related publications or an isolated similarity score. By presenting a classification alongside supporting evidence and an explanation, the engine aims to help researchers identify relevant prior work and formulate questions for further investigation.

The research also provides an opportunity to evaluate the strengths and limitations of combining semantic similarity techniques with generative language models for research-gap assessment. The reported experiments investigate classification accuracy, comparisons with baseline methods, the contribution of LLM reasoning, and processing time.

The component is intended as an assistive research tool. It does not replace a comprehensive literature review, expert assessment, or the judgement of academic supervisors. Its practical significance depends on the relevance of retrieved evidence, the reliability of generated explanations, and the extent to which users can understand and verify its outputs.

1.9 Thesis Structure

This thesis is organized into eight chapters.

Chapter 1 introduces the overall project, the individual component, the research problem, the research gap, the aim, objectives, research questions, scope, and significance.

Chapter 2 reviews relevant literature and existing techniques related to academic literature retrieval, semantic representations, evidence extraction, graph-based analysis, and LLM-assisted research-gap assessment. It synthesizes the limitations of existing approaches and establishes the conceptual basis for the proposed engine.

Chapter 3 describes the research methodology, including the research design, study materials, experimental procedure, evaluation metrics, analysis methods, and considerations relating to responsible AI and reproducibility.

Chapter 4 presents the product and solution design, including the requirements, overall architecture, design alternatives, detailed pipeline design, and relevant professional practices.

Chapter 5 describes the implementation, integration, testing procedures, and verified technical evidence.

Chapter 6 presents the research results, technical outcomes, and validation evidence available for the completed component.

Chapter 7 interprets the findings, compares them with relevant approaches, and discusses the contributions, limitations, and implications of the research.

Chapter 8 concludes the thesis, reviews the achievement of the research objectives, summarizes the individual contributions, and identifies future research directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This chapter reviews research and existing techniques relevant to automated research-gap assessment. The review focuses on five connected areas: academic literature retrieval, semantic text representation, extraction of gap-relevant evidence, graph-based analysis of research relationships, and LLM-based reasoning.

The purpose of the review is to establish the technical background for the proposed engine and examine how existing approaches address different parts of the research-gap identification problem. The review also considers the limitations that motivate combining these capabilities within a single assessment pipeline.

The accompanying research paper and its reference list provide the initial sources for this review. Relevant original publications should be consulted to verify the methods and findings attributed to them. Additional literature should be included where necessary to establish a balanced understanding of the field.

Literature was searched in Google Scholar, IEEE Xplore, the ACL Anthology, arXiv and Semantic Scholar. Search terms combined “research gap detection”, “novelty detection”, “semantic similarity”, “sentence embeddings”, “retrieval-augmented generation”, “topic modelling” and “LLM-as-a-judge”. Sources from [2015/2018] to 2026 were considered, favouring peer-reviewed venues and foundational papers; non-peer-reviewed sources were used only for tools and products, cited from official documentation. This is a targeted narrative review, not a systematic review.

2.2 Domain and Theoretical Foundations

2.2.1 Research-Gap Identification and Idea Validation

A research gap refers to an unresolved question, limitation, insufficiently investigated relationship, or opportunity for further study within a research domain. Identifying such a gap requires understanding the current state of knowledge and examining what existing studies have not adequately addressed.

Research-gap identification is related to, but distinct from, literature discovery. Literature discovery retrieves potentially relevant publications, whereas gap assessment requires interpretation of their contributions and limitations in relation to a particular proposed idea.

The distinction is important because an idea may be related to existing work without duplicating its contribution. A proposed study may extend an established method to a new context, address a limitation in a previous study, or investigate a question that remains unresolved. Conversely, different terminology may conceal substantial conceptual overlap.

The engine therefore treats research-gap assessment as a comparative reasoning task involving both the proposed idea and evidence extracted from relevant literature. Its output is an advisory classification rather than a definitive statement of novelty.

2.2.2 Academic Literature Retrieval

Academic literature retrieval identifies publications relevant to a query or proposed research idea. Traditional retrieval methods often rely on keyword matching and term-weighting techniques. Such methods are useful when the query and documents share important terms, but lexical overlap does not always reflect conceptual similarity.

The system uses a retrieval approach that combines a local FAISS index with external academic sources, including arXiv and Semantic Scholar. This arrangement is intended to support retrieval from locally available documents as well as publications available through external services.

A local index can support efficient similarity searches over a prepared collection, while external sources can expand the range of potentially relevant publications. However, retrieval quality remains dependent on the available corpus, query formulation, indexing process, and source coverage.

Retrieval is therefore an initial stage rather than a complete solution to research-gap identification. The retrieved documents must still be analysed to determine which information is relevant to the proposed research idea.

2.2.3 Semantic Text Representation and Embeddings

Semantic text representation converts text into numerical vectors that can be compared computationally. Transformer-based sentence-embedding models are designed to represent textual meaning in a vector space, allowing related passages to be compared even when they do not use identical words.

The results show the use of `all-MiniLM-L6-v2`, which generates 384-dimensional embeddings. These embeddings are used to represent the research idea and relevant literature evidence.

A common method of measuring the similarity between two embedding vectors is cosine similarity:

$$Cosine\ (A, B) = (A \cdot B) / (|A| \times |B|) \quad \textbf{(2.1)}$$

where (A) and (B) represent the two embedding vectors, (A · B) is their dot product, and (|A|) and (|B|) are their Euclidean norms.

Cosine similarity provides a numerical indication of the directional similarity between vector representations. However, a high score does not prove that two research contributions are equivalent, and a low score does not prove that an idea is novel. The

score must be interpreted in relation to the task, the retrieved evidence, and other information available to the assessment process.

Sentence-BERT and related sentence-embedding research provide relevant foundations for comparing sentence-level and passage-level representations. The original publications should be cited when discussing the model architecture and embedding approach in detail.

2.2.4 Targeted Evidence Extraction

Academic publications contain different types of information across their sections. Abstracts summarize the study, methodology sections explain how it was conducted, and discussion, limitations, and future-work sections may identify unresolved issues or opportunities for further research.

Processing an entire document without distinguishing these roles can introduce information that is only indirectly related to the question being investigated. Targeted evidence extraction seeks to identify passages that are particularly relevant to the assessment task.

The context-aware extraction stage that focuses on sections associated with limitations and future research. It uses extracted passages as evidence for subsequent semantic analysis and LLM reasoning.

The usefulness of this stage depends on the structure and quality of the source documents. Section headings may differ across publishers, and some publications may not clearly separate limitations and future work. Extraction errors or missing sections can therefore reduce the amount or relevance of the evidence supplied to later stages.

Targeted extraction should consequently be evaluated not only by whether the pipeline runs, but also by whether the selected passages contain relevant information and support the resulting explanation.

2.2.5 Graph-Based Semantic Analysis

A graph represents entities as nodes and relationships between them as edges. In a literature-analysis context, nodes may represent a proposal and retrieved papers, while edges represent relationships such as semantic similarity or other defined forms of overlap.

Graph analysis can provide a structural view of the relationships between documents. For example, a paper connected to many other documents may occupy a central position within the retrieved literature network.

The research paper describes a dynamic similarity graph and a blended centrality score combining normalized proposal-to-paper similarity and normalized degree centrality:

Centrality=0.7 Sim_Cosine+0.3 D_cen **(2.2)**

where (S) is normalized proposal similarity and (D) is normalized degree centrality.

The weighting gives greater emphasis to the proposal's direct similarity to a paper while incorporating the paper's position within the graph. The score is intended to support interpretation of the relationships among retrieved papers and the proposal.

However, graph centrality is not a direct measurement of novelty. A highly connected paper may be influential or broadly related to a topic without necessarily resolving the particular problem addressed by a proposed idea. The contribution of graph analysis must therefore be considered in relation to the other stages of the pipeline and the reported ablation results.

2.2.6 Large Language Models and Evidence-Based Reasoning

Large Language Models can interpret natural-language context and generate explanations, classifications, and summaries. In literature analysis, they can be used to compare a proposed idea with retrieved evidence and produce a structured explanation of the relationship.

Retrieval-augmented generation combines retrieved information with a generative model so that responses can be informed by external material. Research on retrieval-augmented generation provides a relevant foundation for understanding how retrieved documents may be incorporated into model outputs.

Nevertheless, an LLM can produce unsupported statements, overlook relevant evidence, or assign inconsistent classifications to borderline cases. Supplying relevant evidence and requiring a structured output may help constrain the task, but it does not eliminate these risks.

The research paper describes a few-shot prompting approach in which semantic signals and extracted passages are provided to an LLM. The model produces one of three classifications: `GAP_ADDRESSING`, `PARTIAL_OVERLAP`, or `HIGH_SIMILARITY`, together with a supporting explanation.

The assessment is intended to assist human judgement. Its classification and explanation must remain open to review, particularly where the retrieved evidence is incomplete or the distinction between extending existing work and proposing a genuinely different contribution is ambiguous.

2.3 Related research

2.3.1 Academic Knowledge Graphs and Literature Discovery

Academic knowledge graphs and scholarly retrieval systems organize relationships between publications, authors, citations, and research topics. The Semantic Scholar

literature graph described by Ammar et al. [1] provides an example of scholarly information represented through structured relationships.

Such systems can help users discover relevant publications and explore connections within an academic domain. However, discovering connected publications is not equivalent to determining whether a particular proposed research idea addresses a limitation identified in those publications.

For this thesis, scholarly retrieval and graph-based literature organization are relevant as supporting techniques. Their relationship to the research problem lies in identifying and structuring the evidence that an idea-level assessment requires.

2.3.2 Research-Gap Extraction

Automated research-gap extraction aims to identify potential areas for further investigation from existing scholarly literature. GapFinder [12] is an example of a system that uses AI to explore research gaps.

Gap extraction can support researchers by highlighting possible areas that remain insufficiently investigated. However, this work differs from corpus-level gap exploration, because it compares a specific, early-stage idea against the limitations and unresolved issues of relevant studies.

This distinction motivates an idea-level assessment process. Instead of only identifying a general research opportunity, the engine attempts to determine how a submitted idea relates to evidence drawn from related publications.

2.3.3 Dense Semantic Retrieval

Dense semantic retrieval represents queries and documents as vectors and compares their positions within an embedding space. Sentence-BERT [3] is a relevant foundation for generating sentence embeddings that can be compared efficiently.

FAISS [4] provides a framework for efficient similarity search over dense vectors. Its use in the reported system supports the retrieval of relevant documents from a prepared local index.

These techniques address different parts of the pipeline: embeddings represent semantic content, while vector-search infrastructure retrieves documents or passages with similar representations. Neither technique independently determines whether a research idea resolves an existing limitation.

2.3.4 Retrieval-Augmented Generation

Retrieval-augmented generation combines retrieved information with generative language models to support responses that draw on external knowledge. Lewis et al. [5] introduced a retrieval-augmented generation framework for knowledge-intensive

NLP tasks, while Izacard and Grave [6] investigated passage retrieval with generative models for open-domain question answering.

These approaches are relevant because the research-gap engine must interpret a proposed idea in the context of existing publications rather than relying solely on a model's internal knowledge.

The reported engine differs from a basic retrieval-and-generation process by emphasizing targeted extraction of gap-relevant sections and incorporating semantic and graph-derived signals before the final classification. Whether these additional stages improve classification accuracy or primarily improve explanation quality is an empirical question examined in the evaluation.

[Citations required: Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," 2020; Izacard and Grave, "Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering," 2021.]

2.3.5 Topic Modelling and Trend Discovery

Topic modelling identifies themes in large corpora. Latent Dirichlet Allocation (LDA) [14] model documents as mixtures of topics, while BERTopic [10] combines transformer embeddings with clustering and class-based TF-IDF to produce more coherent topics. These methods support trend analysis, but they describe what a corpus contains and do not test whether a specific proposal addresses an unresolved limitation. Domain-specific encoders such as SciBERT [2] and document-level models such as SPECTER [13] improve scientific text representation, and are considered as future embedding upgrades in this work.

2.3.6 Synthesis of Related Research

The reviewed approaches contribute complementary capabilities. Scholarly retrieval systems identify potentially relevant publications, semantic embeddings support meaning-oriented comparisons, evidence extraction focuses analysis on relevant parts of documents, graph methods represent relationships, and LLMs provide natural-language reasoning and explanations.

However, the presence of these individual capabilities does not establish that a combined system will reliably determine whether a proposed research idea addresses a research gap. The relationship between similarity, unresolved evidence, and novelty remains complex, particularly when proposed ideas extend existing methods rather than introduce entirely new concepts.

The proposed engine investigates a combination of these techniques in a single pipeline. Its contribution must be assessed using classification outcomes, comparisons with simpler baselines, processing-time measurements, and careful consideration of the limitations of the available benchmark.

2.3.7 LLM Evaluation Reliability

Using an LLM as a classifier or judge introduces known risks. Zheng et al. [8] show that LLM judges can exhibit biases and overconfidence, and RAG surveys [9] describe hallucination and context-selection problems even when retrieval is used. These findings motivate this work's design choices of constraining the LLM to retrieved evidence, using a strict few-shot output schema, and blending the model's self-reported confidence with a signal derived from similarity scores.

2.4 Existing systems products and techniques

Existing academic tools and techniques can be compared according to the type of assistance they provide.

System / Approach	**Main capability**	**Idea-level validation?**	**Limitation for this problem**
Google Scholar / Semantic Scholar [1]	Literature search and literature graph	No	Retrospective; no comparison with a new idea
Connected Papers	Visual citation-graph exploration	No	Maps published work only
Elicit, Consensus	LLM-assisted synthesis of findings	No	Answers research questions; does not critique novelty
GapFinder [12]	Corpus-level research-gap exploration	No	Domain-level gaps, not tied to a submitted idea
BERTopic / LDA [10], [14]	Topic and trend discovery	No	Themes, not unresolved limitations
TF-IDF matching [16]	Keyword overlap	Partial	Fails on lexically different but equivalent concepts
Naive RAG [5]	Retrieved chunks passed to an LLM	Partial	Noisy fixed-size chunks; weaker grounding
Proposed engine	Retrieval + targeted extraction + alignment + LLM reasoning	Yes (advisory)	Small benchmark; depends on retrieval coverage

The research also evaluates a pure TF-IDF baseline and a naive retrieval-augmented generation configuration. These are particularly useful comparisons because they help investigate whether semantic analysis and targeted evidence handling offer benefits beyond keyword matching or unstructured passage retrieval.

2.5 Gap Synthesis and Conceptual Framework

The literature reviewed in this chapter demonstrates that automated research-gap assessment involves several connected tasks. Relevant publications must first be identified, useful evidence must be extracted, and the proposed idea must be compared with that evidence. A final assessment must then interpret these relationships without treating similarity as definitive proof of duplication or novelty.

The conceptual framework for this research combines five stages:

1. **Similarity-guided retrieval:** Identify relevant academic publications from the available local and external sources.
2. **Targeted evidence extraction:** Select passages associated with limitations, unresolved problems, and future research.
3. **Semantic representation:** Convert the proposal and selected evidence into dense vector representations.
4. **Semantic alignment and graph analysis:** Calculate similarity signals and examine relationships among the proposal and retrieved documents.
5. **LLM-based reasoning:** Interpret the supplied evidence and signals to generate a structured classification and explanation.

The framework assumes that each stage provides information that can contribute to the final assessment. Retrieval narrows the literature considered, extraction focuses attention on relevant evidence, semantic representations support comparison, graph analysis adds structural context, and LLM reasoning produces an interpretable output.

These stages also introduce potential sources of error. Irrelevant retrieval results can misdirect the analysis, extraction may omit important passages, similarity measures may not capture the distinction between novelty and overlap, and the language model may generate unsupported or inconsistent conclusions.

Consequently, the framework must be evaluated as an integrated process and through comparisons that investigate the contribution of individual stages. The benchmark and ablation experiments provide initial evidence, while the small dataset limits the conclusions that can be drawn.

The resulting conceptual framework motivates the research methodology presented in Chapter 3 and the system design described in Chapter 4.

3 RESEARCH METHODOLOGY

3.1 Research Design

This research adopts a software-artifact development and evaluation approach consistent with the Design Science Research direction described in the approved project proposal. The approach focuses on addressing an identified practical problem through the design and development of a technological artifact and evaluating its performance against defined objectives.

The artifact is the Similarity-Guided Semantic Analysis and Gap Detection Engine. It is intended to support research-gap assessment by combining literature retrieval, evidence extraction, semantic representation, graph-based analysis, and LLM reasoning.

The research process can be described through the following activities:

1. **Problem identification:** Establish the difficulty of assessing a proposed research idea using related literature and identify the limitations of relying solely on retrieval or similarity scores.
2. **Objective definition:** Define the required capabilities of an evidence-oriented research-gap assessment engine.
3. **Artifact design and development:** Organize the retrieval, extraction, semantic-analysis, and reasoning stages into a coherent pipeline.
4. **Demonstration and evaluation:** Apply the pipeline to labelled research scenarios and compare its outputs with simpler approaches and selected alternative configurations.
5. **Analysis and refinement:** Interpret the observed results, identify limitations, and determine where additional implementation or evaluation work is required.

The accompanying research paper provides the initial experimental baseline for the evaluation described in this thesis. Additional experiments will be included only where their setup and outcomes can be verified.

3.2 Data participants or study materials

The primary evaluation evidence consists of a manually curated benchmark containing 12 scenarios. These scenarios are used to assess the system's ability to classify relationships between proposed research ideas and relevant literature.

The paper reports three classification labels:

- `GAP_ADDRESSING`
- `PARTIAL_OVERLAP`
- `HIGH_SIMILARITY`

The benchmark is used to calculate strict three-class classification accuracy and compare the full pipeline with selected baseline and ablation configurations.

The retrieval environment includes a local FAISS index containing academic papers and external academic retrieval sources, including arXiv and Semantic Scholar. The paper describes a local cache of 22 peer-reviewed papers spanning five NLP domains, alongside dynamic retrieval from external sources.

The benchmark scenarios and the literature used during retrieval serve different purposes. The scenarios provide labelled cases for evaluating classification, while retrieved publications provide evidence for the system's analysis. Their roles should be distinguished when reporting the experimental procedure.

The benchmark comprises 12 manually curated scenarios with reference classifications established by the researcher. Independent inter-rater validation was not conducted; therefore, these classifications serve as the operational reference labels for this evaluation rather than independently verified ground truth. The accuracy measures agreement between the system predictions and these reference classifications. The results should consequently be interpreted as an initial assessment of performance on the selected benchmark, rather than a definitive measure of the system's ability to determine research novelty.

Retrieval used the arXiv API (over 2.4 million open-access articles), the Semantic Scholar Graph API (over 200 million papers) and a local FAISS cache of 22 peer-reviewed papers across five NLP domains. Experiments ran on an Intel Core i7-12650H (10 cores, 2.30 GHz), 16 GB DDR5 RAM and an NVIDIA RTX 4050 Laptop GPU (6 GB); LLM inference was run without GPU acceleration.

3.3 Experimental or study procedure

The evaluation examines the full pipeline and selected alternative configurations. The procedure includes comparing classification outputs against benchmark labels and measuring end-to-end processing time.

The main configurations are summarized below.

Configuration	Purpose
Full system using local LLaMA 3.2	Evaluates the combined retrieval, extraction, semantic-analysis, graph-analysis, and LLM-reasoning pipeline
Groq cloud configuration using OpenAI GPT-OSS 20B	Examines performance with a cloud-based reasoning backend
TF-IDF baseline	Provides a keyword-based comparison
Naive RAG without targeted extraction	Investigates a retrieval-and-generation approach without section-focused evidence extraction
Heuristic-only configuration without LLM reasoning	Examines classification without the LLM reasoning stage

Configuration without blended centrality	Investigates the effect of removing the graph-centrality contribution

Alignment and centrality. The cosine similarity between the proposal embedding and each evidence embedding establishes an asymmetric alignment in which the proposal acts as the hypothesis and prior limitations act as premises. Papers are nodes of a similarity graph whose edges are typed (for example, methodology_similar or gap_related). The blended centrality of each paper is:

$$Centrality = 0.7 \times Sim_Cosine + 0.3 \times D_cen \quad \textbf{(3.1)}$$

where Sim_Cosine is the proposal-to-paper cosine similarity normalised by the maximum observed similarity, and D_cen is the paper's degree centrality normalised by the maximum degree in the graph. The weights favour proposal-centric similarity because a single proposal is assessed against a corpus. A sensitivity analysis varying the proposal weight from 0.5 to 0.9 in steps of 0.1 produced stable classifications, with output-confidence variation below 3%.

Confidence. The final confidence is not the LLM's raw self-report:

$$Conf_final = 0.6 \times Conf_LLM + 0.4 \times Conf_signal \quad \textbf{(3.2)}$$

where Conf_signal is derived from the spread and count of cosine scores across the retrieved evidence passages. A minimum confidence of 0.3 is required for GAP_ADDRESSING, set empirically because proposals below this value produced inconsistent LLM classifications across repeated runs.

Fallback behaviour. If the LLM is unavailable, the system falls back to a cosine-threshold heuristic: HIGH_SIMILARITY when average cosine similarity exceeds 0.75, and GAP_ADDRESSING when it is below 0.35 with fewer than two evidence passages. In cloud deployments where local Ollama inference is unavailable, the fallback chain prioritises the Groq API (GPT-OSS 20B) with the same structured-output contract.

PDF extraction. pdfplumber performs layout-aware parsing first. If text density indicates a scanned document, pytesseract OCR is used, and PyPDF2 is the final standard extraction layer. Target sections ("Limitations", "Future Work") are isolated with deterministic regular expressions.

The evaluation uses the same 12-scenario benchmark as the basis for comparison. The research paper reports classification accuracy for these configurations and average end-to-end latency for the local and cloud configurations.

3.4 Measures and evaluation metrics

3.4.1 Classification Accuracy

Classification accuracy measures the proportion of benchmark scenarios for which the predicted class matches the reference label.

Accuracy = (Number of correct predictions / Total number of predictions) × 100% **(3.3)**

For the benchmark, the denominator is 12 scenarios. Consequently, a single changed prediction alters accuracy by approximately 8.33 percentage points.

Accuracy provides an easily interpretable measure of overall classification performance. However, it does not reveal the types of errors made or whether one class is predicted more reliably than another. The small benchmark size also means that the percentages are sensitive to individual cases.

3.4.2 Baseline and Ablation Comparisons

Baseline comparisons examine the full system against simpler approaches. The TF-IDF and naive RAG configurations provide comparisons with keyword-based matching and retrieval-based generation without targeted extraction.

Ablation experiments investigate the effect of removing a component or capability. The heuristic-only configuration examines the effect of removing LLM reasoning, while the configuration without blended centrality examines the role of the graph-derived score.

These comparisons help investigate the contribution of individual design choices. Their interpretation must account for the small benchmark and any differences in retrieval data or configuration between experiments.

3.4.3 Processing Time

End-to-end latency measures the elapsed time required to complete an analysis request, from the beginning of the pipeline to the generation of the result.

The results show average end-to-end latency for local and cloud reasoning configurations. These measurements provide an indication of the processing cost under the experimental conditions.

Latency comparisons depend on hardware, model configuration, network conditions, retrieval operations, and the number and size of documents processed. The averages should therefore be interpreted in the context of the experimental environment rather than treated as universal performance guarantees.

3.4.4 Additional Metrics

In addition to accuracy, precision, recall and F1-score are computed per class from the per-scenario predictions, and a confusion matrix is reported (Section 6.1). End-to-end latency is reported per configuration, and stage-level latency (retrieval, embedding, graph construction, LLM inference) is recorded by the pipeline's built-in timing diagnostics. Because the Groq configuration varied between runs, results are reported over [n] repeated runs as mean ± standard deviation. Expert agreement is reported in Section 6.3 if the evaluation described there was completed.

3.5 Data analysis

The evaluation uses descriptive comparisons of classification accuracy and processing time across the full system, baselines, and selected ablations.

Classification outcomes are summarized as the number of correct predictions out of the 12 benchmark scenarios and the corresponding percentage. Processing time is reported as average end-to-end latency for the configurations.

The analysis also considers the differences between classification accuracy and explanation quality. In particular, the paper reports that the full system and naive RAG achieved the same number of correct classifications, while the full system produced more targeted, evidence-grounded explanations.

This distinction is important because equal classification accuracy does not establish that the systems provide equivalent explanations, but qualitative differences in explanation quality should not be treated as a measured improvement unless they have been evaluated using a clearly defined procedure.

Similarly, the ablation without graph centrality did not demonstrate a consistent classification-accuracy improvement from including the blended centrality score. The score may still contribute to structural interpretation, but this should be distinguished from an improvement in classification accuracy.

No statistical significance, broad generalization, or expert agreement should be claimed unless supported by additional appropriate data and analysis.

3.6 Ethics privacy security and responsible AI

The component processes research ideas that may contain unpublished findings, proposed methods, or other information that users consider confidential. Protecting this information is therefore an important design and operational consideration.

The project proposal identifies confidentiality and privacy as non-functional requirements. It states that research ideas should not be permanently stored or shared without authorization. These requirements should guide implementation and

deployment decisions, but their presence in the proposal does not by itself establish that the running system satisfies them.

The system handles submitted proposals as follows. Proposal text is held in memory during processing and is not stored permanently by the AI component; only the final assessment is saved by the core application in MongoDB. Retrieval queries derived from the proposal are sent to arXiv and Semantic Scholar. With local inference (Ollama/LLaMA 3.2), no proposal text is sent to an LLM provider. With the Groq configuration, the proposal and extracted evidence are sent to the provider and are subject to its data-handling terms [confirm and cite Groq's current retention policy]; for confidential proposals, the local configuration is therefore recommended. All outputs are advisory, and a HIGH_SIMILARITY result must not be treated as evidence of plagiarism.

Responsible use also requires recognizing that automated research-gap assessments can be incorrect. A high-similarity classification should not automatically be treated as evidence of plagiarism, and a gap-addressing classification should not be taken as proof of originality.

The engine is consequently positioned as a decision-support tool. Its outputs should provide relevant evidence and explanations to facilitate review by researchers or academic supervisors, who remain responsible for final academic judgements.

3.7 Reproducibility and research artefacts

Reproducibility requires sufficient documentation of the software configuration, benchmark, experimental procedure, and evaluation results for the findings to be understood and, where possible, independently reproduced.

The relevant artefacts for this research include:

- The benchmark scenarios and reference labels.
- The retrieval configuration and sources used.
- The embedding model and vector-search configuration.
- The evidence-extraction rules.
- The LLM model and prompt configuration.
- The graph-analysis and classification settings.
- The evaluation procedure and recorded outputs.
- The software environment and relevant dependency versions.

The research paper identifies `all-MiniLM-L6-v2` as the embedding model and describes local LLaMA 3.2 and a Groq cloud configuration using OpenAI GPT-OSS 20B. These details provide part of the baseline configuration, but additional implementation and environment details should be verified against the relevant code and experiment records.

Where the current implementation differs from the paper's configuration, the difference should be recorded explicitly. This distinction is necessary to avoid presenting results from one configuration as though they were produced by another.

4 PRODUCT AND SOLUTION DESIGNA

4.1 Requirements

The requirements for the Semantic Analysis and Gap Detection Engine are derived from the problem described in Chapter 1 and the intended functionality set out in the project proposal.

They are divided into functional requirements, which define what the engine should do, and non-functional requirements, which define important qualities and constraints.

4.1.1 Functional Requirements

ID	Requirement	Description
FR1	Research idea input	Accept a research idea or proposal description for analysis through the larger system's interface.
FR2	Relevant literature retrieval	Obtain a set of relevant academic publications for the submitted idea.
FR3	Evidence extraction	Extract relevant evidence from available document sections, including limitations, future work, methodology, discussion, and other relevant passages.
FR4	Semantic representation	Generate vector representations of the proposal and selected literature evidence.
FR5	Semantic alignment	Calculate similarity signals between the proposed idea and the retrieved evidence.
FR6	Relationship analysis	Use semantic and graph-derived information to support analysis of relationships among the proposal and retrieved publications.
FR7	LLM-based assessment	Use an LLM to interpret the available evidence and produce a structured assessment.
FR8	Classification output	Produce a classification corresponding to `GAP_ADDRESSING`, `PARTIAL_OVERLAP`, or `HIGH_SIMILARITY`.
FR9	Explanation generation	Provide an explanation that relates the classification to the evidence and analysis signals available to the system.

These requirements describe the intended capabilities of the component. The degree to which each is implemented and verified should be established in Chapters 5 and 6.

4.1.2 Non-Functional Requirements

ID	Requirement	Design consideration
NFR1	Security	Protect research ideas and prevent unauthorized access to confidential information.

NFR2	Privacy	Minimize unnecessary persistence or disclosure of submitted ideas and verify the handling of data sent to external services.
NFR3	Performance	Complete analysis within a response time suitable for the intended research-support workflow.
NFR4	Usability	Provide understandable classifications and explanations that users can interpret.
NFR5	Reliability	Handle retrieval failures, document-processing problems, and model errors in a controlled manner.
NFR6	Scalability	Support changes in the number of requests and the size of the available literature collection.
NFR7	Maintainability	Allow relevant models, retrieval sources, and analysis components to be updated without unnecessary redesign of the entire system.
NFR8	Interpretability	Provide supporting evidence and explanations rather than returning only an unexplained similarity score or classification.

These non-functional requirements originate from the project requirements and the intended use of the component. They should be treated as design objectives until implementation and testing evidence demonstrates the extent to which they have been satisfied.

4.2 Component architecture within the complete solution

The Semantic Analysis and Gap Detection Engine operates as one component of the broader AI-powered research-support system. Its role is to perform detailed semantic assessment after relevant literature has been identified for a submitted research idea.

The project proposal describes the component as an independent Python-based service with an API entry point and a sequential analysis pipeline. At a conceptual level, the architecture consists of five main stages: similarity-guided retrieval, structural evidence extraction, semantic representation and alignment, graph-based analysis, and LLM reasoning.

The engine receives a research idea and the information required to identify relevant literature. It then processes selected documents, generates semantic signals, and supplies the available evidence and analysis results to the reasoning stage. The final assessment contains a classification and supporting explanation intended for delivery to the larger application.

The component's boundaries are important because literature discovery and research-gap assessment serve different purposes. The broader platform is responsible for the overall user workflow, while the engine focuses on interpreting the relationship between the proposed idea and retrieved literature. (Figure 4.1: Overall system context diagram)

The diagram should show the research-support platform, the Semantic Analysis and Gap Detection Engine, the other relevant modules, and the direction of information flow. Include only integrations supported by the actual system design or implementation. If a connection is proposed but not yet implemented, identify it as an intended integration.

4.3 Design alternatives and decisions

4.3.1 Retrieval-Guided Analysis

One possible approach to research-gap assessment would be to analyse the proposed idea directly using an LLM without retrieving supporting publications. However, this would not provide a sufficiently explicit connection between the assessment and the current literature.

The selected approach retrieves relevant publications and uses their contents as evidence for the subsequent analysis. The research paper describes a combination of a local FAISS index and external academic retrieval services. This arrangement is intended to support access to a prepared local collection and a broader set of publications.

The effectiveness of this decision depends on retrieval relevance and coverage. A relevant publication that is not retrieved cannot contribute to the assessment, and irrelevant publications can introduce misleading context.

4.3.2 Targeted Evidence Extraction

A second design decision concerns the amount of document text passed to the analysis stages. Processing full documents or arbitrary text chunks may include information that is not directly relevant to identifying research gaps.

The selected approach emphasizes passages from sections associated with limitations, future work, and other relevant evidence. This is intended to focus the later reasoning stage on material that can help explain whether a proposed idea addresses an unresolved issue.

The trade-off is that document structures vary, and relevant evidence may appear outside expected sections. The extraction process therefore remains dependent on the quality and structure of available documents.

4.3.3 Dense Semantic Representation

Keyword-based methods such as TF-IDF provide a relatively simple way to measure textual overlap. However, they may fail to capture relationships expressed using different terminology.

The system uses dense embeddings to support meaning-oriented comparison between the proposal and literature evidence. The `all-MiniLM-L6-v2` model is the embedding model identified in the research paper.

This choice supports a computational representation of semantic relationships, but the resulting similarity values do not directly establish novelty. They must be interpreted alongside the extracted evidence and reasoning output.

4.3.4 Graph-Based Analysis

The design incorporates graph-based analysis to represent relationships between the proposal and retrieved publications. A blended score combines normalized proposal similarity and normalized degree centrality, with weights of 0.7 and 0.3, respectively.

This design adds information about the structural position of papers in the similarity graph. However, graph centrality does not independently determine whether a proposal addresses a research gap. The evaluation without blended centrality also indicates that this design choice should not be assumed to improve classification accuracy.

The graph component is therefore best understood as an additional source of structural information whose practical contribution must be assessed separately from the accuracy of the final classification.

4.3.5 Embedding Model Selection

all-MiniLM-L6-v2 [3], [11] was selected over scientific-domain encoders such as SciBERT [2] and SPECTER [13] because it produces compact 384-dimensional vectors, runs with low latency without heavy GPU infrastructure, and was sufficient for the retrieval and alignment signals needed. Domain-specific models are expected to improve precision on highly technical text and are listed as future work.

4.3.6 LLM-Based Reasoning

The final stage uses an LLM to interpret the retrieved evidence and semantic signals. The research paper describes few-shot prompting and a structured classification output.

The use of an LLM allows the system to produce a natural-language explanation and consider relationships that may not be captured by a single numerical threshold. However, model outputs can vary, and explanations may contain unsupported conclusions.

For this reason, the classification should remain advisory and should be reviewed alongside the supporting evidence. The ablation results, in which accuracy decreases when LLM reasoning is removed, provide an empirical reason to investigate the contribution of this stage.

4.3.7 LLM Backend Selection

A locally hosted LLaMA 3.2 [7] was chosen as the primary backend because it keeps proposal text on the user's machine and has no per-request cost, at the price of higher latency (22.9 s) and a smaller model. A Groq-hosted GPT-OSS 20B backend was added for cloud deployments, reducing latency to 17.2 s but sending text to a third party and showing more conservative classification on borderline cases. Both share one output schema so the backend can be switched through configuration.

4.3.8 Extraction Strategy

Deterministic regular-expression section extraction was preferred over the trained sequence labeller (BiLSTM-CRF) considered at proposal stage because it requires no labelled training data, is transparent and fast, and was sufficient to isolate Limitations and Future Work sections. Its limitation is sensitivity to non-standard headings.

4.4 Detailed design

4.4.1 Stage 1: Similarity-Guided Retrieval and Preprocessing

The pipeline begins by identifying academic publications relevant to the submitted research idea. The research paper describes retrieval from a local FAISS index and external academic sources, including arXiv and Semantic Scholar.

The purpose of this stage is to reduce the literature considered by later stages to a subset of potentially relevant publications. Preprocessing prepares the retrieved material for document analysis.

The retrieval stage can influence every subsequent stage. If relevant publications are omitted, their evidence cannot be used; if unrelated publications are retrieved, they may introduce misleading signals. Retrieval quality is therefore a dependency of the complete pipeline.

4.4.2 Stage 2: Context-Aware Evidence Extraction

The second stage identifies text passages that may contain information relevant to research-gap assessment. These include limitations, future-work recommendations, discussion, and other sections containing relevant research evidence.

The extracted passages are intended to reduce irrelevant context and provide the semantic-analysis and reasoning stages with information more directly related to unresolved problems.

The stage depends on the availability and structure of the source documents. PDF extraction errors, inconsistent section headings, and missing text can affect the quality of the resulting evidence.

[Insert Figure 4.2: Document-processing and evidence-extraction flow.]

The figure should illustrate how retrieved documents are processed and how gap-relevant passages are selected. Include specific extraction libraries or fallback behaviour only after confirming them against the implementation.

4.4.3 Stage 3: Semantic Representation

The third stage transforms the research idea and extracted evidence into dense vector representations. According to the research paper, the system uses `all-MiniLM-L6-v2` to produce 384-dimensional embeddings.

The vectors provide a common representation for comparing the proposal with relevant evidence. Their similarity can be calculated using cosine similarity, which provides a numerical signal for the next stage of analysis.

The semantic representation stage is designed to capture relationships that may not be evident from exact word matching. Nevertheless, similarity values are signals for assessment rather than definitive measures of novelty.

4.4.4 Stage 4: Semantic Alignment and Graph Analysis

The fourth stage examines the semantic relationship between the proposed idea and retrieved publications. The system combines proposal-to-paper similarity with graph-based information.

The blended centrality score defined in Equation (3.1) is computed for each retrieved paper and passed to the reasoning stage as a structural signal.

The score provides a combined view of direct similarity and graph structure. The resulting signals are made available to the reasoning stage as contextual information for the final assessment.

The interpretation of these signals must account for the fact that strong similarity may indicate shared subject matter rather than duplication, while weak similarity does not establish that an idea is original.

4.4.5 Stage 5: LLM Reasoning and Output Generation

The final stage receives extracted evidence and semantic-analysis signals. The research paper describes few-shot prompting of a local LLaMA 3.2 model, with an alternative Groq cloud configuration using OpenAI GPT-OSS 20B.

The model is expected to produce a structured assessment using one of the three classification labels:

- `GAP_ADDRESSING`
- `PARTIAL_OVERLAP`
- `HIGH_SIMILARITY`

The result also includes a supporting explanation and a confidence-related score as described in the paper. The score should not be interpreted as a calibrated probability unless calibration has been demonstrated through an appropriate evaluation.

The output is intended to explain the relationship between the proposal and the available literature evidence. Where the retrieved material is insufficient or ambiguous, the assessment should be interpreted cautiously and should not replace human academic judgement.

[Insert Figure 4.3: Five-stage end-to-end pipeline.]

The diagram should connect all five stages and show the flow from a submitted idea to retrieved publications, extracted evidence, semantic and graph signals, and the final classification with explanation.

4.5 Standards and professional practice

The design follows ISO/IEC 25010 quality characteristics (performance efficiency, reliability, security, maintainability), the requirements practice of ISO/IEC/IEEE 29148, and basic OWASP guidance for API and secret handling (credentials supplied through environment variables, not source code). Citations follow IEEE style. External services are used under their published terms: the arXiv and Semantic Scholar APIs, and the licences of all-MiniLM-L6-v2 and LLaMA 3.2 [confirm each licence and note it here].

4.5.1 Maintainability and Modularity

Separating retrieval, document processing, semantic representation, graph analysis, and reasoning into distinct stages can make the system easier to understand and maintain. It also provides a basis for evaluating individual stages and changing components when necessary.

However, the benefits of modularity depend on the implementation maintaining clear interfaces and manageable dependencies. The actual level of modularity should be verified from the source code and described in Chapter 5.

4.5.2 Data Protection and Confidentiality

Research ideas may contain unpublished intellectual property. The system should therefore minimize unnecessary persistence, control access to submitted data, and clearly document any external service that receives user-provided text.

The project proposal identifies privacy and security as requirements. The design should be evaluated against these requirements, including the handling of logs, database records, API requests, and hosted-model interactions.

4.5.3 Responsible AI and Human Oversight

The engine produces assessments that may influence how researchers perceive the originality of a proposal. Incorrect classifications could discourage a useful idea or create unjustified confidence that an idea is novel.

The output should therefore be treated as advisory. Users should be able to examine the evidence behind the assessment, and final decisions should remain with researchers and academic supervisors.

The system should also distinguish between measured similarity and a conclusion about research novelty. These are related but different concepts, and the design should avoid presenting a numerical score as definitive proof.

4.5.4 Evaluation and Traceability

The classification, evidence, and relevant analysis signals should be documented sufficiently to support evaluation and debugging. Experimental results should be associated with the model and configuration that produced them.

This is particularly important when local and cloud LLM configurations or different embedding models are compared. Results from different configurations should not be combined without documenting the relevant differences.

The thesis evaluation should maintain a clear distinction between the baseline results reported in the research paper and additional results obtained from later implementation changes.

4.5.5 Licensing and External Dependencies

The component depends on external models, libraries, academic retrieval services, and potentially hosted LLM providers. Their licences, terms of use, rate limits, and data-handling conditions may affect deployment and continued operation.

A complete assessment should verify the licences and conditions relevant to the actual dependencies used. This section should document confirmed findings rather than assume that all models and services permit unrestricted commercial or institutional use.

The design presented in this chapter establishes the intended structure of the engine. Chapter 5 examines how these design elements are implemented and verified in the available software.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The Similarity-Guided Semantic Analysis and Gap Detection Engine combines document processing, semantic representation, similarity analysis, and language-model reasoning to support research-gap assessment. The technologies described in the accompanying research paper provide the baseline for this implementation chapter.

The implementation uses Python for the analysis pipeline, `all-MiniLM-L6-v2` for semantic embeddings, FAISS for local vector similarity search, and academic retrieval sources including arXiv and Semantic Scholar. The document-processing approach includes PDF text extraction and alternative extraction mechanisms for handling different document formats or extraction failures. The reasoning stage uses a local LLaMA 3.2 configuration, with a cloud alternative using OpenAI GPT-OSS 20B through Groq.

Table 5.1: Technologies Used

Technology	**Role in the component**
Python	Implementation of the analysis pipeline
FAISS	Local vector similarity search
`all-MiniLM-L6-v2`	Generation of 384-dimensional semantic embeddings
arXiv	External academic literature retrieval
Semantic Scholar	External academic literature retrieval
PDF-processing tools	Extraction of text from academic documents
Local LLaMA 3.2	Local LLM-based reasoning configuration
Groq with OpenAI GPT-OSS 20B	Alternative cloud-based reasoning configuration

5.2 Component implementation

The component is organized around a sequential pipeline in which each stage supplies information to the next. The stages are designed to progressively transform a research idea into an evidence-supported assessment of its relationship to existing literature.

The implementation described in the research paper consists of literature retrieval, targeted evidence extraction, semantic embedding and similarity analysis, graph-based analysis, and LLM reasoning. This section describes their implementation at a functional level. Detailed source-code references, configuration names, and implementation-specific behaviour should be added after checking the corresponding code.

5.2.1 Retrieval and Document Processing

The first stage retrieves publications that may be relevant to the submitted research idea. The implementation combines a local FAISS index with external academic retrieval through arXiv and Semantic Scholar.

The local index supports similarity-based retrieval over an available collection of academic documents. External retrieval provides an additional source of potentially relevant publications. Retrieved documents are then prepared for subsequent analysis.

Document processing is necessary because academic publications may be available as PDF files rather than directly usable text. The research paper describes a PDF extraction approach involving `pdfplumber`, `pytesseract`, and `PyPDF2` as part of its document-processing and fallback strategy.

The purpose of this stage is to make the contents of retrieved publications available to later stages while handling differences in document structure and extraction quality. The effectiveness of the stage depends on whether the extracted text preserves the relevant content of the original document.

5.2.2 Embedding and Similarity Implementation

After documents have been retrieved and processed, the engine generates semantic representations of the proposed research idea and relevant literature text.

The baseline described in the research paper uses `all-MiniLM-L6-v2`, which produces 384-dimensional embeddings. These representations allow the engine to compare texts according to their positions in a vector space rather than relying solely on the presence of matching words.

Cosine similarity provides a numerical measure of the similarity between two vector representations. The resulting scores are used as signals for the subsequent semantic-alignment and reasoning stages.

The embedding and similarity stages are important because the same concept may be expressed using different terminology in different publications. Dense representations can support comparisons across such wording differences, although they cannot independently establish whether two research contributions are equivalent.

5.2.3 LLM Reasoning and Output Handling

The reasoning stage interprets the evidence extracted from relevant publications together with the available semantic-analysis signals. The research paper describes few-shot prompting to guide the model toward a structured classification.

The baseline uses local LLaMA 3.2, with a cloud alternative using OpenAI GPT-OSS 20B through Groq. The model is expected to return one of three classifications:

- `GAP_ADDRESSING`
- `PARTIAL_OVERLAP`
- `HIGH_SIMILARITY`

The classification is accompanied by an explanation and a confidence-related score as described in the paper. The score is an indicator produced by the system and should not be described as a statistically calibrated probability unless a calibration study supports that interpretation.

Structured output handling is important because downstream application components need to interpret the assessment consistently. Invalid or incomplete model responses can interfere with this process, so the implementation should be examined for output validation and error-handling behaviour.

5.3 Integration with the Other Components

The Semantic Analysis and Gap Detection Engine forms part of a broader research-support platform that also includes the Integrity and Governance Module, Supervisor Recommendation Module, and Workspace with NLP Assistance Module.

At the system level, the engine's purpose is to assess a proposed research idea against relevant academic literature. Its output can support a wider research workflow by providing a classification and explanation that may be reviewed alongside other information produced by the platform.

Integration should be discussed at two distinct levels. **Conceptual integration** describes how the component is intended to work with the broader platform. **Verified integration** describes communication that has actually been implemented and tested.

For example, a conceptual workflow may involve a user submitting a research idea through the application, the engine processing it, and the resulting assessment being returned for display. This workflow should only be described as operational if the corresponding interface, API contract, and end-to-end behaviour have been verified.

Table 5.2: Integration responsibilities

Component	Potential relationship with the gap detection engine	Evidence required
Integrity and Governance Module	May provide complementary information about research integrity or similarity	Confirm the actual data exchange and workflow
Supervisor Recommendation Module	May use research-topic information to support supervisor matching	Confirm whether the engine's output is consumed by this module

Workspace with NLP Assistance	May provide the user-facing research workflow through which an idea is submitted or an assessment is viewed	Confirm the interface, API calls, and result presentation
Overall application	Coordinates the user workflow and presents the component's result	Confirm the implemented integration path

This distinction prevents proposed interactions from being presented as completed features. If the engine was evaluated independently of the other modules, state that explicitly and describe system-level integration as incomplete or outside the evaluation scope, as appropriate.

Figure 5.1: Component integration diagram. The diagram should show only verified connections as solid lines. Planned or conceptual connections can be shown differently and clearly labelled.

5.4 Verification and software testing (Testing Strategy)

Testing is required to establish whether the engine's individual stages and complete pipeline behave as intended. The test categories below describe a suitable structure for reporting verification. Actual outcomes must be based on recorded test execution rather than expected behaviour.

5.4.1 Component testing

Component testing examines individual stages in isolation.

Relevant checks include whether retrieved documents can be processed, whether embeddings are generated with the expected dimensions, whether similarity calculations return valid numerical values, and whether the reasoning stage produces an output that can be parsed into the expected classification structure.

These checks help identify faults within individual stages before evaluating the complete pipeline.

Test area	Test action	Expected behaviour
Document processing	Process a supported academic PDF	Text is extracted for subsequent analysis
Embedding generation	Encode a sample research idea and passage	Embeddings are generated with the expected dimension
Similarity calculation	Compare two valid embeddings	A valid similarity value is returned
Classification parsing	Supply a valid structured model response	The classification and explanation are extracted

Invalid model response	Supply or simulate an incomplete response	The system handles the invalid response according to its implemented error-handling strategy

5.4.2 Pipeline testing

Pipeline testing evaluates the stages together using a research idea and the literature retrieved for that idea.

A pipeline test should confirm that the input reaches the retrieval stage, documents are processed, relevant evidence is selected, semantic signals are generated, and the reasoning stage returns the expected output structure.

An end-to-end test should also check whether failures in one stage are handled appropriately and whether the final response is delivered to the caller or user interface.

Where the system depends on external academic services or a hosted LLM, the testing record should identify whether those services were available during the test. A successful test using cached or mock data should not be presented as proof that the external-service integration is working.

5.4.3 Regression testing

Regression testing checks whether changes to the implementation cause previously working behaviour to fail.

For this component, relevant regression cases include representative research ideas from the labelled benchmark, expected output structure, embedding-dimension compatibility, retrieval behaviour, and classification handling.

Because the component uses an LLM, exact text matching may not be appropriate for every output. Regression criteria should distinguish between properties that must remain stable, such as valid labels and output structure, and properties that may vary, such as the wording of an explanation.

Where regression results differ from previous results, the configuration, model, retrieval data, and relevant test case should be recorded to help determine the cause.

5.4.4 Configuration testing

Configuration testing verifies that the engine behaves as expected under the model and service settings used in an experiment.

Relevant settings include the embedding model, retrieval sources, LLM backend, prompt configuration, graph-analysis weights, and other parameters that affect the classification process.

This testing is particularly important when comparing local and cloud reasoning configurations. If the embedding model or retrieval collection changes at the same time as the LLM provider, the results cannot automatically be attributed to the provider change alone.

5.4.5 Performance testing

Performance testing examines how long the engine takes to complete an analysis request under specified conditions.

The research paper reports average end-to-end latency for the local and cloud configurations. These figures provide the baseline performance results discussed in Chapter 6.

Additional performance tests should record the number of test runs, the hardware and service conditions, the size of the input, and whether retrieval or model responses were cached. These details help distinguish a representative measurement from a single observation.

5.5 Reported Testing Evidence

The accompanying research paper provides benchmark results for classification accuracy, selected baseline comparisons, an ablation without LLM reasoning, and average end-to-end latency. These results constitute the current documented evaluation evidence for the thesis.

They should be distinguished from software verification evidence such as unit-test outputs, API responses, screenshots, or regression-test logs. Benchmark accuracy indicates how the system performed on the evaluated scenarios; it does not by itself prove that every software component behaves correctly under all conditions.

Table 5.4: Evidence available for the thesis

Evidence category	What it establishes	Reporting requirement
Benchmark accuracy	Classification performance on the 12 scenarios	Report the configuration and denominator
Baseline comparison	Relative accuracy of the full system and selected simpler approaches	State the comparison conditions and limitations
Ablation result	Observed performance when a selected component is removed	Identify the component removed and the resulting configuration
Latency measurement	Average processing time under tested conditions	State whether the measurement was local or cloud-based

Software test output	Behaviour of specific functions or integration paths	Include only tests actually executed and retain their logs or screenshots

5.6 Deployment and Operation

The engine's operation depends on the availability of its software environment, retrieval sources, document-processing dependencies, embedding model, and reasoning backend.

A local reasoning configuration and a hosted reasoning configuration have different operational considerations. Local inference depends on the available computing resources and model installation, while cloud inference depends on network access, provider availability, API credentials, rate limits, and the provider's data-handling terms.

Deployment documentation should identify the actual operating environment and configuration used for the evaluated system. It should also explain how required credentials are supplied and protected, how failures are reported, and whether research ideas or analysis outputs are persisted.

The thesis should not claim that the system is production-ready, publicly deployed, scalable, or compliant with a particular privacy standard unless this has been demonstrated through implementation evidence and appropriate testing.

If deployment remains provisional, describe the current execution method and the remaining work needed for reliable operation. Further deployment improvements may include clearer configuration management, dependency pinning, service-health checks, error logging that avoids sensitive content, and documented recovery procedures.

6 RESULTS AND VALIDATION

6.1 Research results

The evaluation uses 12 manually curated scenarios to examine the classification performance of the proposed engine. The evaluation compares the full system with alternative configurations, including a TF-IDF baseline, a naive retrieval-augmented generation (RAG) configuration without targeted evidence extraction, and a heuristic-only configuration without LLM reasoning.

6.1.1 Classification Accuracy

The following classification accuracy results have been produced during evaluations.

Table 6.1: Classification accuracy

Configuration	**Backend**	**Accuracy**	**Avg. latency**
Full system (local)	LLaMA 3.2	58.3% (7/12)	22.9 s
Full system (cloud)	GPT-OSS 20B via Groq	41.7–58.3%	17.2 s
Baseline 1: TF-IDF	Deterministic	25.0% (3/12)	1.1 ms
Variant A: no context extraction (naive RAG)	LLaMA 3.2	58.3% (7/12)	15.7 s
Variant B: no blended centrality	LLaMA 3.2	50.0–58.3%	21.9–33.5 s
Variant C: no LLM (heuristic)	Cosine thresholds	33.3% (4/12)	6.7 s

The full local system correctly classified seven of the 12 scenarios, corresponding to an accuracy of 58.3%. The naive RAG configuration achieved the same accuracy, while the TF-IDF baseline achieved 25.0% and the heuristic-only configuration achieved 33.3%.

These findings indicate that the full system outperformed the TF-IDF and heuristic-only configurations on this benchmark. However, its accuracy was equal to that of the naive RAG configuration. Consequently, the classification results alone do not demonstrate that every additional stage of the proposed pipeline improves accuracy.

The cloud configuration produced results within the range of 41.7% to 58.3%. This variation does not support a conclusion that the cloud configuration consistently outperformed the local configuration. A more detailed comparison would require the exact model settings, per-run results, and experimental conditions.

The benchmark contains only 12 scenarios, meaning that one additional correct or incorrect prediction changes the overall accuracy by approximately 8.33 percentage points. The results should therefore be interpreted as preliminary evidence rather than a definitive estimate of general performance.

6.1.2 Effect of Targeted Evidence Extraction

The evaluation compares the full system with a naive RAG configuration that does not use the same targeted evidence-extraction approach. Both configurations achieved a classification accuracy of 58.3%.

The research paper reports that targeted evidence extraction was more beneficial for the grounding and quality of explanations than for classification accuracy. This suggests that selecting relevant passages from limitations and future-work sections may help the reasoning stage produce explanations that are more directly connected to the literature, even when the final classification does not change.

However, this distinction must be interpreted carefully. The available accuracy results establish that the two configurations achieved the same accuracy, but a claim of improved explanation quality should be supported by the qualitative evaluation or examples documented in the research paper. It should not be interpreted as a statistically demonstrated improvement unless a suitable evaluation was conducted.

6.1.3 Effect of LLM Reasoning

The heuristic-only configuration, which removes LLM reasoning, achieved 33.3% accuracy, compared with 58.3% for the full local system.

The difference is 25 percentage points, equivalent to three additional correct classifications out of 12 scenarios. This result suggests that LLM reasoning contributed to classification performance under the experimental conditions.

Nevertheless, the comparison does not establish that LLM reasoning will provide the same benefit across larger or different datasets. The result depends on the benchmark, retrieval evidence, model configuration, and other pipeline stages. It should therefore be interpreted as evidence of a contribution in the experiment rather than a general guarantee.

The heuristic variant predicted PARTIAL_OVERLAP for 11 of the 12 scenarios. This matched four of the six true PARTIAL_OVERLAP labels but failed on every GAP_ADDRESSING and HIGH_SIMILARITY case. Expanding the retrieval pool to include live arXiv results (40 papers instead of 20) raised average similarity scores and pushed the fixed thresholds towards the middle class. This shows that cosine distributions for open gaps and saturated work overlap, and that separating them requires reasoning over the evidence, which the LLM stage provides.

6.1.4 Effect of Graph Centrality

The research paper reports an ablation experiment that removes the graph-centrality contribution from the analysis. Removing graph centrality did not materially change the classification accuracy.

This finding indicates that the blended graph-centrality score did not produce a measurable classification-accuracy improvement on the small benchmark used in the study.

The result does not necessarily mean that graph analysis has no value. Graph-derived information may still help represent relationships among retrieved publications or support explanation and interpretation. However, the experiment does not establish a clear accuracy benefit from including centrality in the final classification process.

The distinction between a component's conceptual usefulness and its measured effect on a specific metric is important when interpreting this result.

6.2 Product and technical results

The research paper describes a five-stage pipeline that combines literature retrieval, targeted evidence extraction, semantic representation, graph-based analysis, and LLM reasoning. The technical configuration uses `all-MiniLM-L6-v2` for 384-dimensional embeddings and supports local and cloud reasoning configurations.

The component's intended output is a structured classification supported by an explanation. The three labels represent different relationships between a proposed research idea and the retrieved literature:

- `GAP_ADDRESSING` indicates that the proposal is assessed as addressing a potential gap.
- `PARTIAL_OVERLAP` indicates that the proposal has some overlap with existing work.
- `HIGH_SIMILARITY` indicates substantial similarity to existing research.

These labels are decision-support categories, not definitive findings about academic originality. The reliability of each classification depends on the retrieved evidence and the model's interpretation of it.

The evaluation also demonstrates the feasibility of comparing a combined pipeline against simpler alternatives and examining the effects of removing selected stages. However, the results do not establish that the engine can reliably assess all research domains or all forms of novelty.

Table 6.2: Stage-level latency (local configuration)

Stage	**Time**
Retrieval	1.3–4.2 s
Embedding	3.8–4.2 s
Graph construction	< 10 ms
LLM reasoning and novelty	~13–21 s

Retrieval outliers. For a rare-domain proposal (low-resource Dhivehi speech recognition, TC4A), retrieval took 110.7 s because the system exhausted several API pages searching for scarce literature. A re-run with a warm cache (TC4B) completed in 15.6 s, which supports cache priming and on-demand domain crawling for obscure topics.

6.3 User or stakeholder validation

The evaluation conducted in this study assesses the system's classification performance against labelled benchmark scenarios, alongside comparisons with baseline approaches, an ablation experiment, and processing-time measurements. These experiments provide technical evidence of the system's behaviour under the evaluated conditions.

Technical evaluation is distinct from user or stakeholder validation. The benchmark results assess classification performance against the reference labels used in the experiments. They do not, by themselves, establish whether researchers or academic supervisors consider the system's explanations useful, its outputs trustworthy, or its workflow suitable for practical research activities.

A formal stakeholder evaluation was outside the scope of the experiments conducted for this study. Accordingly, usability, explanation quality, and practical value from the perspective of researchers or academic supervisors have not been independently established. This limitation does not negate the implementation or the experimental findings; it defines the extent of the conclusions that can be drawn from the evaluation.

Future work could involve researchers or academic supervisors reviewing the evidence supporting individual classifications and assessing the relevance, traceability, and usefulness of the generated explanations. Such an evaluation would require a defined protocol, appropriate participants, explicit assessment criteria, and suitable ethical consideration.

The present study therefore establishes an initial technical evaluation of the implemented engine, while independent expert assessment and end-user validation remain areas for further investigation.

6.4 Integration results

The engine is intended to form part of a broader AI-powered research-support platform. Its role within that platform is to provide a semantic assessment of a proposed idea in relation to relevant academic literature.

However, the research paper's classification and latency results do not, by themselves, establish that every integration with the Integrity and Governance Module, Supervisor Recommendation Module, or Workspace with NLP Assistance has been completed and tested.

Integration results should therefore be reported according to the actual state of the implementation. If the engine has been tested independently, describe the test boundary and state that broader integration was outside the experiment. If end-to-end integration has been tested, include the relevant request, response, and test evidence.

6.5 Summary of findings

The results provide initial evidence that the combined engine can classify research-idea scenarios and can be evaluated against simpler configurations.

The principal findings are:

1. The full local system achieved 58.3% accuracy on the 12-scenario benchmark.
2. The naive RAG configuration achieved the same accuracy, indicating that the additional pipeline stages did not improve the overall classification-accuracy figure on this benchmark.
3. The full system outperformed the TF-IDF baseline and heuristic-only configuration.
4. The cloud configuration varied between 41.7% and 58.3%, so consistent superiority over the local configuration cannot be established.
5. Targeted extraction was reported to contribute more clearly to explanation grounding than to classification accuracy.
6. Removing graph centrality did not materially change classification accuracy in the experiment.
7. The small benchmark limits generalization, and the results do not establish independent expert validation or calibrated confidence scores.

These findings form the basis for the discussion in Chapter 7. They indicate the potential value of combining evidence extraction and LLM reasoning while highlighting the need for broader evaluation and more detailed validation of individual components.

7 DISCUSSION

7.1 Interpretation of findings

The evaluation indicates that the proposed engine can perform research-gap classification on the scenarios included in the benchmark. The full local system achieved 58.3% accuracy, outperforming the TF-IDF baseline and the heuristic-only configuration.

The results also show that the performance of the complete pipeline should not be interpreted as proof that every stage contributes equally to classification accuracy. The naive RAG configuration achieved the same accuracy as the full system, while removing graph centrality did not materially change the classification result.

These findings suggest that the value of a multi-stage architecture should be examined at more than one level. Overall accuracy is important, but it may not capture differences in the relevance of retrieved evidence, the grounding of explanations, or the usefulness of the output to a researcher.

The benefit of targeted extraction in explanation grounding is therefore relevant to the intended purpose of the engine. However, explanation quality must be evaluated separately from classification accuracy. If an explanation is to be treated as a measurable advantage, the evaluation method and supporting examples should be documented clearly.

The results also indicate that the choice of reasoning configuration matters. The cloud configuration varied within the accuracy range, rather than consistently outperforming the local configuration. Model choice should consequently be considered alongside retrieval quality, prompt design, processing time, reliability, and data-handling requirements.

Classification is more challenging in borderline cases where a proposed research idea may partially overlap with existing work while still addressing an unresolved aspect of the topic. In such cases, differences between the system's prediction and the benchmark reference classification may arise from model limitations, ambiguity in the case, or limitations in the reference-label assignment process. The current evaluation does not independently establish which factor accounts for each disagreement. These cases indicate that clearer classification criteria and evaluation by independent academic reviewers would strengthen future validation.

Overall, the evidence supports treating the engine as a research-support prototype whose behaviour requires further evaluation, rather than as an authoritative system for determining originality.

7.2 Comparison with literature baselines and existing solutions

The evaluation includes comparisons with TF-IDF, naive RAG, and a heuristic-only configuration. These comparisons help assess whether the combined approach performs differently from simpler alternatives on the same small benchmark.

The full system achieved 58.3% accuracy, compared with 25.0% for TF-IDF. This result suggests that the complete pipeline performed better than the keyword-based baseline under the experimental conditions.

The heuristic-only configuration achieved 33.3%, which was lower than the full system's 58.3%. This difference is consistent with the possibility that LLM reasoning helps interpret retrieved evidence and semantic signals. However, because the evaluation uses only 12 scenarios, the result should not be generalized without additional experiments.

The comparison with naive RAG is particularly important. Both configurations achieved 58.3% accuracy, so the results do not establish an improvement in classification accuracy from targeted extraction and the other additional stages as a whole. The observation about explanation grounding provides a separate potential benefit, but it should not be substituted for evidence of improved classification.

The graph-centrality ablation similarly suggests that the blended score did not materially change classification accuracy on the benchmark. This finding raises a design question about the complexity and operational cost of graph analysis relative to its measurable contribution. Further evaluation would be required to determine whether graph analysis improves other outcomes or benefits scenarios not represented in the benchmark.

The TF-IDF baseline’s 25.0% accuracy [16] reflects its inability to treat lexically different but semantically equivalent terms as the same concept: it flagged non-identical terminology as high overlap and could not recognize equivalence such as GAT versus GNN. The heuristic’s 33.3% shows that similarity statistics alone cannot separate open gaps from saturated domains. The naive RAG variant [5] matched the full system’s accuracy (58.3%), but its explanations relied on incidental keyword overlap in fixed 500-word chunks instead of targeted limitation passages.

These comparisons are specific to the configurations and scenarios. They do not constitute a comprehensive evaluation against every academic discovery platform, research-gap detection system, or commercial research assistant.

7.3 Research contribution

The research contributes an implemented approach to comparing early-stage research ideas with evidence retrieved from academic literature. It combines several complementary capabilities into a single pipeline: similarity-guided retrieval, targeted

evidence extraction, dense semantic representations, graph-based analysis, and LLM reasoning.

A second contribution is the use of structured classification outputs to represent the relationship between a proposed idea and existing research. The three categories provide a consistent framework for communicating an assessment while recognizing that the underlying relationship may be uncertain or partial.

The experimental contribution is an initial comparison of the combined system with simpler configurations. The benchmark results show differences between the full system, TF-IDF, and heuristic-only approaches, while the RAG comparison and graph-centrality ablation identify areas where additional complexity did not produce a clear classification-accuracy improvement.

The findings therefore contribute not only an architectural approach but also an initial assessment of its limitations. In particular, they highlight the need to distinguish overall classification accuracy from explanation quality and to avoid assuming that every additional analysis stage improves every evaluation metric.

The contribution is limited by the size and composition of the benchmark. The work should be presented as a proof of concept and an initial evaluation, not as conclusive evidence that automated research-gap identification has been solved.

7.4 Product contribution

From a product perspective, the engine is intended to assist researchers by connecting a proposed research idea with relevant evidence from existing publications. The structured output aims to help users examine possible overlap, identify unresolved issues, and determine where further literature review is needed.

The component could be useful in an undergraduate or early-stage research workflow, where users may benefit from a structured starting point for discussing a proposal with a supervisor. Its value depends on whether the retrieved evidence is relevant, the explanations are traceable, and the limitations are communicated clearly.

The engine is not a substitute for academic databases, comprehensive literature reviews, or expert judgement. A classification can be affected by incomplete retrieval, document-extraction errors, model behaviour, and the difficulty of distinguishing a genuinely new contribution from an extension of existing work.

Accordingly, the product contribution is best characterized as decision support. Its output should guide further investigation rather than determine whether a proposal is original, acceptable, or ready for submission.

7.5 Limitations and threats to validity

7.5.1 Small Evaluation Dataset

The evaluation uses only 12 manually curated scenarios. With this sample size, a single prediction changes the accuracy by approximately 8.33 percentage points. The results are therefore sensitive to individual cases and may not represent performance on a broader dataset.

The benchmark's composition may also affect the measured outcomes. If the scenarios do not adequately represent the range of research domains, writing styles, and degrees of overlap encountered in practice, the results may not generalize to those settings.

7.5.2 Limited Evidence of Generalization

The findings concern the 12 scenarios and system configurations evaluated in the accompanying research paper. Although the results provide an initial assessment of the engine's classification performance, they do not establish how it would perform across the full range of academic disciplines, document types, or stages of research development.

Further evaluation should include a larger and more diverse set of research scenarios, with clearly documented reference classifications. Evaluation across different retrieval collections and reasoning configurations would also help establish how consistently the engine performs under changing experimental conditions.

7.5.3 Similarity Is Not Equivalent to Novelty

Semantic similarity measures relationships between text representations, but research novelty depends on the meaning and contribution of the underlying ideas. Two studies may use similar terminology while addressing different questions, and two conceptually overlapping studies may use different terminology.

The engine's similarity signals must therefore be interpreted alongside extracted evidence and human review. Neither a high nor a low similarity score independently establishes originality or duplication.

7.5.4 LLM Variability and Unsupported Outputs

LLM-based reasoning can be affected by prompt wording, model selection, retrieved context, and provider behaviour. The variation in cloud accuracy illustrates that configurations may not produce identical results.

A generated explanation may also contain a conclusion that is not adequately supported by the retrieved literature. Structured prompts and evidence-oriented inputs can help guide the model, but they do not eliminate the need for validation.

7.5.5 Graph-Centrality Contribution

The ablation without graph centrality did not materially change classification accuracy. This limits the evidence for a direct classification benefit from the blended score.

Additional evaluation could investigate whether graph analysis contributes to explanation quality, ranking, or other outcomes not captured by overall accuracy. Until then, the thesis should avoid claiming that centrality improves classification performance.

7.5.6 Confidence-Score Interpretation

The system produces a confidence-related score, but the baseline does not establish that the score is statistically calibrated. It should therefore be treated as a system-generated indicator rather than a verified probability that the classification is correct.

Calibration would require comparing predicted confidence with observed correctness over an adequate set of labelled examples.

7.5.7 Lack of Established Expert Validation

The benchmark results do not establish independent expert validation or a formal user study. Consequently, the practical usefulness, perceived trustworthiness, and suitability of the explanations for academic decision-making remain insufficiently established by the current baseline.

7.5.8 Implementation and Configuration Changes

The implementation may evolve after the publication of the research paper. Results obtained with a newer embedding model, LLM provider, prompt, retrieval index, or routing configuration cannot automatically be treated as equivalent to the baseline.

A reproducible thesis must identify which code and configuration produced each result. Unvalidated changes should be described as exploratory work or future directions, not as established improvements.

7.6 Implications

The findings have several implications for the further development of automated research-gap assessment.

First, evaluation should measure more than the final classification label. Explanation grounding, evidence relevance, consistency, and the ability to identify missing or insufficient evidence are also important for a system intended to assist academic judgement.

Second, additional components should be justified through evidence. The graph-centrality ablation shows why architectural complexity should not automatically be interpreted as improved performance. Each stage should be assessed according to the outcome it is intended to improve.

Third, benchmark design is central to the credibility of the evaluation. A larger and more diverse set of scenarios, with documented reference labels and a transparent annotation process, would provide a stronger basis for comparison.

Fourth, responsible use requires human oversight. The engine should make it clear that its output is advisory and that final conclusions about research novelty require examination of the relevant literature and the proposed contribution.

Finally, changes to models and retrieval configurations should be tested systematically. This would make it easier to distinguish genuine improvements from variation caused by different models, prompts, datasets, or operating conditions.

Taken together, these implications support continued development of the engine while emphasizing the importance of rigorous evaluation, evidence traceability, and transparent limitations.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research investigated the development and evaluation of a Similarity-Guided Semantic Analysis and Gap Detection Engine intended to support the assessment of early-stage research ideas against relevant academic literature.

The work addresses a challenge that is not fully resolved by literature retrieval alone: determining how a proposed idea relates to the limitations, unresolved problems, and future research directions reported in existing studies. To address this challenge, the research paper describes a five-stage pipeline combining similarity-guided retrieval, targeted evidence extraction, semantic representation, graph-based analysis, and LLM reasoning.

The evaluation used 12 manually curated scenarios and compared the full system with selected baseline and ablation configurations. The full local system achieved 58.3% accuracy, outperforming the TF-IDF and heuristic-only configurations. However, the naive RAG configuration achieved the same accuracy as the full system, and removing graph centrality did not materially change classification accuracy.

These findings suggest that the combined approach is a useful direction for investigation, but they do not establish that every stage improves classification performance. The small benchmark, variation in cloud results, and absence of established expert validation limit the conclusions that can be drawn.

The engine should therefore be understood as an assistive research tool rather than a definitive novelty detector. Its output can help researchers identify relevant evidence and investigate potential overlap, but academic decisions must remain grounded in literature review and human judgement.

8.2 Achievement of objectives

The objectives established in Chapter 1 provide a basis for reviewing the work completed and the evidence available.

Table 8.1: Objective achievement assessment

Objective	**Status**	**Evidence**
1. Evidence extraction	Achieved	Section 5.2.1; T1–T4
2. Semantic representation	Achieved	Section 5.2.2; T5–T6
3. Semantic alignment	Achieved	Sections 3.3, 4.4.4 (Table 6.2)
4. LLM reasoning	Achieved	Sections 5.2.3, 6.1; 58.3% local accuracy
5. Evaluation	Partially achieved	12-scenario benchmark, baselines, ablations, latency.

The assessment distinguishes evidence of a capability being included in the system from evidence that the capability achieves its intended purpose reliably. In particular, the small benchmark provides initial evaluation evidence but does not establish broad generalization.

Before final submission, each objective should be checked against the actual implementation and any additional verified experiments. If a feature was only planned or remained incomplete, the table should be revised to reflect that status.

8.3 Individual research and product contributions

The individual research contribution is the investigation of a pipeline that combines semantic comparison with evidence extraction and LLM reasoning for research-gap assessment. The evaluation also provides an initial comparison with simpler approaches and examines the effect of removing selected components.

The product contribution is a component intended to support researchers in examining how a proposed research idea relates to existing publications. Its structured output is designed to provide an accessible starting point for further literature review and discussion with academic supervisors.

These contributions are limited by the available evaluation evidence. The results do not establish that the engine can conclusively identify all research gaps, determine absolute novelty, or replace expert assessment.

The main contribution is therefore a prototype approach and its initial empirical evaluation, including evidence of both promising behaviour and limitations that inform further development.

8.4 Future directions

Several areas could strengthen the reliability and practical value of the engine.

8.4.1 Expand the Evaluation Benchmark

The benchmark should be expanded beyond 12 scenarios to cover more research domains, writing styles, and types of relationship between proposed ideas and existing work. Reference labels should be documented, and the annotation process should be transparent.

A larger benchmark would provide a more informative assessment of classification accuracy and allow additional metrics, such as precision, recall, and F1-score, to be calculated where appropriate.

8.4.2 Conduct Expert Evaluation

A formal evaluation involving researchers or academic supervisors could assess whether the retrieved evidence is relevant, whether explanations are adequately grounded, and whether the classifications are useful in practice.

Such an evaluation should define the participants, assessment criteria, procedure, and analysis method in advance. It should be conducted with appropriate ethical consideration and should not be presented as completed until the study has taken place.

8.4.3 Evaluate Explanation Quality

Future work should evaluate the relationship between generated explanations and the source evidence. Possible criteria include evidence relevance, factual support, coverage of important limitations, and consistency between the explanation and the assigned classification.

This would help establish whether targeted evidence extraction improves explanation quality beyond what can be inferred from classification accuracy alone.

8.4.4 Investigate Embedding and LLM Configurations

Alternative embedding models and LLM providers could be evaluated under a consistent experimental protocol. The same benchmark, retrieval conditions, and clearly documented settings should be used wherever possible so that comparisons are interpretable.

Exploratory configurations involving alternative embeddings, local/cloud routing, and different prompts should remain separate from the baseline until their behaviour has been verified and their results recorded.

8.4.5 Improve Calibration and Uncertainty Handling

Future work could examine whether the system's confidence-related score corresponds to observed correctness. If sufficient labelled data becomes available, calibration methods and uncertainty-aware output could be investigated.

The system could also be improved to indicate when evidence is insufficient or conflicting, rather than forcing a strong conclusion in every case.

8.4.6 Examine the Contribution of Graph Analysis

Further experiments could investigate whether graph-based analysis improves ranking, evidence organization, or explanation quality even when it does not materially affect classification accuracy.

Such experiments should define the intended benefit in advance and measure it directly.

8.4.7 Improve Reliability, Privacy and Deployment

Further engineering work could focus on consistent configuration management, robust error handling, dependency management, and operational monitoring. Privacy and security improvements should include verifying how research ideas are handled by local services, external retrieval sources, hosted LLM providers, logs, and persistent storage.

Deployment claims should be supported by testing in the target environment, including service availability, error recovery, response time, and appropriate data-handling controls.

8.4.8 Explore Broader Research Workflows

The component could be evaluated as part of the wider research-support platform to determine whether its outputs are useful when combined with research integrity checks, supervisor recommendation, and NLP-assisted workspace functionality.

This should begin with verified integration and usability testing rather than assuming that the value of the standalone engine automatically transfers to the full platform.

In conclusion, the work establishes an initial foundation for semantic research-gap assessment and identifies concrete areas for further evaluation. Larger benchmarks, transparent evidence handling, expert assessment, and reproducible experiments are necessary steps toward determining its reliability and practical usefulness.

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GLOSSARY

- **Asymmetric alignment:** comparing a proposal (the hypothesis) against prior limitations (the premises) rather than measuring symmetric document similarity.
- **Ablation:** an experiment that removes one component to measure its contribution.
- **Context-aware extraction:** isolating specific sections (Limitations, Future Work) instead of fixed-size text chunks.
- **Degree centrality:** the number of connections a node has in a graph, normalised by the maximum degree.
- **Few-shot prompting:** giving an LLM worked examples in the prompt to fix the output format and reasoning style.
- **Grounding:** constraining an LLM's answer to supplied evidence to limit hallucination.
- **Human-in-the-loop:** a design in which a person makes the final decision.

APPENDICES

Appendix A Research instruments or extended results

- A.1 Benchmark scenarios: a table with ID, proposal title or text, reference label, predicted label (local), predicted label (Groq), and latency for all 12 scenarios.
- A.2 Expert evaluation instrument: consent statement and Likert items: (1) The retrieved papers are relevant to the proposal. (2) The identified gap is correct. (3) The explanation is clear. (4) The explanation is supported by the cited evidence. (5) I would find this useful when reviewing a proposal. (1 = strongly disagree … 5 = strongly agree.)
- A.3 Risk analysis (from your proposal): LLM hallucination → structured, evidence-grounded prompts; PDF extraction errors → rule-based section detection and OCR fallback; scarce labelled data → expert validation and synthetic cases; high compute cost → top-N retrieval and batching; training-data bias → diverse datasets and checks.

Appendix B Academic integrity and AI use disclosure

In line with the module policy, generative AI was used as an assistive tool during the preparation of this thesis. All research design decisions, system implementation, experiments, results and conclusions are verified.

Table B.1: AI tools used

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic)	Suggesting the chapter structure and content, drafting and language editing of text, proper grammar and formatting.	Text checked against the project proposal, ICAC paper, source code and experiment logs.	Chapters 1-9 (drafting and editing)
GitHub Copilot	Code suggestions, debugging support	Code reviewed and tested	Section 5.4